

LAB - 7 CONFIGURE AND CONNECT THE COMPUTER IN LAN

Step -1 Select the 05 - PCs, 01- Printer, 01- Router, 01- Switch and 01- Server By using Drag and Drop Step -2 Configure IP address to Router as 192.168.1.100 Double Click on Router and Select port status check box on . (for which Ethernet switch connected)

Step -3 Configure Server IP address 192.168.1.101 and connect to switch by Straight Through cable. Click on Server, Select Desktop, Select IP Configuration

Step -4 Connect all 05 PCs to switch using Straight Through cable and configure IP address as 192.168.1.1 to 192.168.1.5

Step -5 Give gateway IP as 192.168.1.100 and DNS IP 192.168.1.101 for all 05 PCs Click on PC, Click on Desktop, Click on IP configuration

Step -6 Configure Printer IP 192.168.1.102 and connect to switch using Straight Through cable.

Step -7 Ping all PCs from Server and Printer Click in any one PC, Click on Desktop, Click on Command Prompt and ping any pc from this PC By giving command - > ping IP address(ex 192.168.1.1)

LAB - 8 BLOCK THE WEBSITE USING “WINDOWS DEFENDER FIREWALL” IN WINDOWS 10

Step 1: Launch the Control Panel on your computer.

Step 2: Select “Windows Defender Firewall” followed by “Advanced Settings” on the leftside pane.

Step 3: Right-click on “Outbound Rules” from the menu on the left and select “New Rule.”

Step 4: When a new window pops up, select the “Custom” option followed by “Next.”

Step 5: On the next window, select “All programs” and again select “Next.”

Step 6: Select the “ These IP addresses ” option under “Which remote IP addresses does this rule apply to?” and click next

Step 7: Open the Command Prompt as Administrator by entering “CMD” into the search box.

Step 8 : Enter “nslookup www.facebook.com” and press the Enter button.

Step 9: Click on “Add” and enter the IP addresses you want to block. Then select “Next.”

Step 10 : Make sure to choose the “Block the connection” option and click on “Next.”

Step 11: Choose whether the rule applies to Domain, Private, or Public. You can also select all three.

Step 12 : Select “Next,” add a name or description for this rule, and select “Finish” to complete the action.

Step 13: Finish Check for Blocked website

LAB - 9 SHARE THE FOLDER IN A SYSTEM AND ACCESS THE FILES OF THAT FOLDER FROM OTHER SYSTEM USING IP ADDRESS

Step -1 Create Lan Server Configuration (2- PC's, 1- Server, 1- Switch)

Step -2 IP Address Configuration for Server and PC's

Step -3 For both PC give the DNS server IP

Step -4 Click on Server Click on Services Click on FTP Give the User name and Password (username is – admin, password – admin) Give the permission for admin (Write, Read, Delete, Rename, List) Next Click on add

Step -5 From PC1 Create text file (double click on PC) Click on Desktop Click on Text Editor Click on File > New Type text (Share the file in a system and access the files from other system or server using IP address) Click on Save (Give file name as test1.txt)

Step -6 In the same PC click on command prompt Check the file by entering C: \> dir C:\> ftp 192.168.1.1 (Enter the IP address of Server to connect Server) Give Username and Password ftp> put test1.txt (File transferred from PC1 to Server) ftp> dir (Check the file is transferred to server by giving dir command) test1.txt

Step -7 Select PC2 Click on Desktop Click on Command Prompt C:\> dir Test1.txt no such file will be there in PC2 C:\> ftp 192.168.1.1 (Enter the IP address of Server to connect Server) Give Username and Password ftp> get test1.txt (File transferred from Server to PC2) ftp> dir (Check the file is transferred to PC2) test1.txt

LAB -10 SHARE THE PRINTER IN A NETWORK, AND TAKE A PRINT FROM OTHER PC

Step -1 Select the 02 - PCs, 01- Printer and 01 -Switch By using Drag and Drop

Step -2 Configure IP address to Printer as 192.168.1.3 Double click on printer, Click on Config, Select FastEthernet0, Give IPV4 address. And connect to switch using Straight Through cable.

Step -3 Connect all 02 PCs to switch using Straight Through cable and configure IP address as 192.168.1.1 and 192.168.1.2 Click on PC, Click on Desktop, Click on IP configuration

Step -4 Ping Printer from any PC Click in any one PC, Click on Desktop, Click on Command Prompt and ping printer from any PC By giving command - > ping IP address(ex 192.168.1.3)

LAB -11 CONFIGURE OF WIFI HOTSPOT, AND CONNECT OTHER DEVICES (MOBILE / LAPTOP)

Step -1 Select 02 - PCs, 01- Printer, 01- Laptop, 01- server, 01- smart phone and 01- HomeRouter By using Drag and Drop

Step -2 Configure IP address to of Router → Double click on Router → Click on LAN (give router IP address 192.168.1.10) → Click on Wireless2.4G → Click on WPA-PSK → Give PSK pass Phrase (Password 12345678) → Change the SSID name default to KLEBCA

Step - 3 Make Wired PC to Wireless PC → Double Click on PC select Physical → Click on WMP300N → Off the PC Change Network LAN port to Wireless LAN port by drag and drop

Step - 4 Configure wireless PC → Click on desktop → Click on DHCP → Next click on config → Click on wireless0 → Click on WPA-PSK and give PSK PASS Phrase (12345678) → Change SSID default to KLEBCA
PC will get connect to router

Step -5 Repeat Step 3rd to make Laptop, Printer, Server and Smartphone wireless

Now Wifi connection is ready to ping

Step - 6 Ping Printer from any Devices Click in any device, Click on Desktop, Click on Command Prompt and ping any pc from printer By giving command - > Ping IP address OR Ping By Realtime or Simulation

LAB -12 CONFIGURE OF SWITCHES 1. Configure password (For Login to Switch) Enable for switch configuration steps

Step - 1 Switch> EN or Enable

Step - 2 Switch#Config t / configuration terminal

Step -3 Switch(config)#line con 0

Step -4 Switch(config-line)#password 123456

Step -5 Switch(config-line)#login

Step -6 Switch(config-line)# Exit

Step -7 Switch(config)#Exit

Step -8 Switch# Exit

2. Configure password for configuration switch Step - 1 Switch> EN or Enable Step - 2 Switch# Config t / configuration terminal Step -3 Switch(config)# enable secret 12345678 Step -4 Switch(config)# Exit Step -5 Switch# Exit To check for configuration password (i.e Login to Switch) First need to enter console password User Access Verification Password: (Login password) Switch> En /Enable Password: (Configuration password i.e Secrete password) Switch#

3. Configure Switch hostname as KLEBCA Step -1 Switch# configure t Step -2 Switch(config)# hostname KLEBCA Step -3 KLEBCA(config)#

4. Configure the message of the day as -----
----- - "Well Come
to KLEBCA CISCO Switch Configuration" -----
----- - Step -1

KLEBCA(config)#banner motd # -----
----- "Well Come to KLEBCA CISCO
Switch Configuration" -----
-----# Step - 2 KLEBCA(config)# Exit (To
check the message exit from the configuration) Step - 3 KLEBCA# Exit

LAB -16 CONFIGURE OF VPN USING PACKET TRACER / GNS3

Step - 1 : 02 PC's, 03 Routers Step - 2 : Connect 01 PC to Router1 and other to Router 3 Step - 3 : Configure IP for all PCs PC 1- 192.168.1.2 Gateway IP 192.168.1.1 PC2 – 192.168.2.2 Gateway IP 192.168.2.1

Step - 4 Router 0 Configuration Fast Ethernet 0/0 – 192.168.1.1 Fast Ethernet 0/1 – 1.0.0.2 Step - 5 Router 1 configuration Fast Ethernet 0/0 – 1.0.0.1 Fast Ethernet 0/1 – 2.0.0.1 Step - 6 Router 2 configuration Fast Ethernet 0/0 – 192.168.2.1 Fast Ethernet 0/1 – 2.0.0.2 Step -7 Router 0 Click on Config - > Click on Static - > Add Network - 0.0.0.0 Mask - 0.0.0.0 Next Hop - 1.0.0.1 Step -8 Router 2 Click on Config - > Click on Static - > Add Network - 0.0.0.0 Mask - 0.0.0.0 Next Hop - 2.0.0.1

Step – 9 Click on Router 0 Exit Router#ping 2.0.0.2 Router# config t Router(config)#interface tunnel 1 Router(config-if)#ip address 172.16.1.1 255.255.0.0 Router(config-if)#tunnel source fa 0/1 Router(config-if)#tunnel destination 2.0.0.2 Router(config-if)#no shut

Step -10 Router 2 Router(config)#interface tunnel 2 ip address 172.16.1.2 255.255.0.0 Router(config-if)#tunnel source fa 0/1 Router(config-if)#tunnel destination 1.0.0.2 Router(config-if)#exit

Step -11 Router 0 Click on Config - > Click on Static - > Add Network - 192.168.2.0 Mask - 255.255.255.0 Next Hop – 172.16.1.2 Step -12

Router 2 Click on Config - > Click on Static - > Add Network - 192.168.1.0 Mask - 255.255.255.0 Next Hop – 172.16.1.1 Step – 13 Ping from PC1 – ping 192.168.2.2 Tracert 192.168.2.2